

Creative Personalization Command Center

Business Problem Brief

Publishers increasingly say their goal is that no two viewers should have the same experience—but their current creative and delivery systems can't support that ambition. They have rich first-party data and audience segments across web, app, and email, yet most users still see the same homepage hero, newsletter banner, and in-app promos, with personalization limited mainly to which modules appear rather than how they look and feel.

Producing segment-specific imagery and short-form video is too slow and manual, requiring bespoke briefs, design cycles, and per-channel adaptation, so teams default to a small set of generic brand assets reused across all cohorts. Experimentation is similarly constrained, testing a few creatives against the entire audience instead of systematically learning what works for each segment.

At the same time, any move toward large-scale creative generation raises hard questions about governance: how to guarantee that every generated asset respects brand guidelines, rights and licensing constraints, regional rules, and safety policies; how to ensure that audience segments, content metadata, and model outputs are all using the same, high-quality definitions; and how to trace which data and models were used for which experiences. Without a scalable, governed data and model foundation—where audience, content, and creative assets are managed in a single, trusted catalog—publishers can't safely automate or personalize creative at the depth they envision, leaving engagement and subscription growth well below their potential.

Questions To Answer/ Actions to Perform

Audience Segments

- Define high-value audience segments whose behaviors, interests, and lifecycle stage should materially change the creative they experience.

- Identify which traits within each segment should influence the creative brief, such as topic affinity, subscription propensity, churn risk, device usage, or engagement style.

Creative Search & Retrieval

- Search a governed repository of existing images and videos to find the best base creative assets for a given segment, placement, content type, or campaign objective.
- Retrieve assets along with their metadata, rights constraints, prior performance, and approved usage contexts so generation starts from the right foundation.

Creative Generation

- Use the audience segment plus text instructions to generate new image and video creatives tailored to the target segment and placement.
- Produce multiple variants per segment so teams can explore different concepts, tones, and visual treatments rather than relying on a single output.

Creative Modification & Adaptation

- Resize and reformat creatives for different channels and placements, such as homepage hero, app tile, newsletter banner, or social unit.
- Perform image edits such as inpainting and outpainting to extend, clean up, or adapt existing assets.
- Perform similar video modifications, such as reframing, extending, cropping, swapping opening frames, or generating alternate edits for different placements.

Lineage, Governance & Reuse

- Maintain full lineage for every creative variant, including which base asset it came from, which segment and prompt were used, which model generated or edited it, and which transformations were applied.
- Ensure that all generated and modified assets remain discoverable, auditable, and reusable across teams and channels.

Policy, Rights & Safety Checks

- Automatically evaluate generated and modified creatives against brand guidelines, rights and licensing constraints, regional usage rules, and safety policies before they move forward.
- Flag or block assets that are off-brand, non-compliant, unsafe, or outside their approved usage scope, so only eligible creatives are tested and deployed.

Synthetic Audience Evaluation

- Run the generated creative variants against synthetic audiences representing the same target segment to estimate how that segment is likely to respond before going live.
- Rate each variant across parameters such as propensity to click, expected dwell time, subscription-start propensity, relevance, clarity, fatigue risk, and brand fit.
- Rank variants within each segment and placement so the highest-scoring creatives can be selected for deployment or further refinement.

Selection & Activation

- Choose the best-performing creative variants for each segment and channel based on synthetic-audience scoring, policy checks, and business goals.
- Push approved creatives and their metadata into downstream systems such as Adobe Experience Manager, Adobe Target, a CMS, DAM, or onsite personalization platform for activation.

Continuous Improvement

- Compare real-world results against synthetic predictions, and use those learnings to improve prompts, segment definitions, asset retrieval, model performance, and future creative generation.
- Capture winning creative patterns and playbooks so teams can reuse proven approaches instead of starting from scratch each time.

Data Foundation

Audience & Experience Catalog

Audience Segment Reference

Fields:

- Audience_Segment_ID
- Segment_Name (e.g., News_Junkies_Heavy, Sports_Superfans_CTV, Churn_Risk_Families)
- Segment_Type (Demographic, Behavioral, Lifecycle, Value_Tier)
- Data_Source (CDP, DMP, 1P_Data_Platform)
- Segment_Definition (rule / model ID, feature summary)
- Eligibility_Flags (Can_See_Upsell, Can_See_Ads, Age_Restricted)

Purpose:

Defines the audience cohorts that drive creative generation, testing, and targeting across owned surfaces.

Experience & Placement Catalog

Fields:

- Experience_ID
- Placement_Type (Homepage_Hero, Onsite_Module, App_Promo_Tile, Newsletter_Banner)
- Surface (Web, Mobile_App, CTV_App, Email)
- Dimensions_Spec (Width, Height, Aspect_Ratio)
- Content_Context (News, Sports, Kids, Entertainment, Subscription_Upsell)
- Personalization_Flag (Generic, Segment_Specific, Experiment)

Purpose:

Defines where creatives will run, how they should be formatted, and which placements are eligible for personalization and testing.

Creative Asset Foundation

Base Asset Catalog

Fields:

- Asset_ID
- Asset_Type (Image, Video)
- Source_Type (Studio_Provided, In_House_Creative, Stock, Library_Asset)
- Storage_Location / URI
- Format (PNG, JPG, MP4, etc.)
- Dimensions / Duration
- Content_Tags (Topic, Genre, Brand, Franchise, Seasonality)
- Rights_Profile_ID
- Approved_Usage_Contexts (Web, App, Email, CTV, Social)
- Brand_Guardrail_Flags

Purpose:

Serves as the governed repository of approved base creative assets that generation and editing workflows can start from.

Generated / Modified Creative Assets

Fields:

- Creative_Asset_ID
- Parent_Asset_ID (base asset or prior version)
- Asset_Type (Image, Video)
- Audience_Segment_ID

- Experience_ID
- Prompt_Text / Creative_Instructions
- Generation_Method (Text_To_Image, Image_To_Image, Text_To_Video, Video_Edit)
- Model_ID / Endpoint_ID
- Transformations_Applied (Resize, Crop, Inpainting, Outpainting, Reframe, Extend, etc.)
- Output_URI
- Created_Timestamp
- Version_Status (Draft, Approved, Rejected, Archived)

Purpose:

Stores the generated and edited creatives tied to a target segment and placement, along with the key business and technical metadata needed by the application.

Note:

Unity Catalog provides the deeper lineage, auditability, and governance across the underlying tables, models, jobs, and pipelines, so the schema only needs lightweight reference fields such as parent asset and model ID rather than a separate lineage model.

Policy, Rights & Safety

Rights & Policy Profiles

Fields:

- Rights_Profile_ID
- Allowed_Territories
- Allowed_Channels (Web, App, CTV, Email, Social)
- Expiration_Date
- IP_Owner / Licensor
- Usage_Restrictions (No_Derivatives, No_AI_Modification, Talent_Restrictions)

Purpose:

Defines where and how a creative asset can legally and contractually be used.

Safety & Brand Evaluation

Fields:

- Creative_Asset_ID
- Check_Run_ID

- Policy_Engine_Version
- Safety_Score
- Brand_Compliance_Score
- Policy_Violation_Codes
- Allowed_For_Synthetic_Testing_Flag
- Allowed_For_Production_Use_Flag

Purpose:

Captures whether each generated or modified creative passes brand, rights, and policy checks before testing or activation.

Content & Offer Metadata

Content / Offer Reference

Fields:

- Content_ID
- Content_Type (Article, Episode, Collection, Subscription_Offer, Sponsored_Unit)
- Title / Name
- Content_Genre / Category (News, Sports, Kids, Drama, etc.)
- Editorial_Priority (High, Medium, Low)
- Tentpole_Flag (Yes/No, Event_Name)

Purpose:

Defines the content or offer being promoted by the creative.

Creative–Content Mapping

Fields:

- Creative_Asset_ID
- Content_ID
- Experience_ID
- Intended_Objective (Engagement, Subscription_Start, Sponsored_Click, Reengagement)

Purpose:

Associates each creative asset with the content, module, or offer it is designed to promote.

Synthetic Audience & Evaluation Data

Synthetic Audience Definition

Fields:

- Synthetic_Audience_ID
- Audience_Segment_ID
- Generation_Method (Simulation_Model_ID, Training_Data_Snapshot)
- Size (Number_of_Synthetic_Users)
- Feature_Summary (Behavioral_Profile, Device_Mix, Value_Tier)

Purpose:

Defines the synthetic audiences used to evaluate how each target segment is likely to respond to creative variants.

Creative Evaluation Runs

Fields:

- Evaluation_Run_ID
- Synthetic_Audience_ID
- Creative_Asset_ID
- Experience_ID
- Scenario_Context
- Predicted_Click_Propensity
- Predicted_Dwell_Time
- Predicted_Subscription_Start_Propensity
- Predicted_Fatigue_Risk
- Predicted_Brand_Fit_Score
- Evaluation_Timestamp

Purpose:

Stores synthetic audience scoring outputs used to rank and select creative variants before deployment.

Activation & Downstream Delivery

Activation Mapping

Fields:

- Activation_ID
- Creative_Asset_ID
- Audience_Segment_ID
- Experience_ID
- Downstream_System (Adobe_Experience_Manager, Adobe_Target, CMS, Personalization_Engine)
- External_Reference_Key
- Activation_Status (Pending, Live, Paused, Ended)
- Activation_Start_Timestamp
- Activation_End_Timestamp

Purpose:

Tracks which approved creatives have been pushed into downstream systems for which segments and placements.

Data Quality Requirements

ID Consistency & Join Keys

Standardized Audience_Segment_ID, Experience_ID, and Creative_Asset_ID across: segmentation tables, content/offer metadata, creative catalogs, synthetic audience definitions, and activation mappings.

Consistent Content_ID and Rights_Profile_ID usage so base assets, generated assets, and policy records join cleanly without ambiguous or duplicate keys.

Asset & Record Hygiene

De-duplicate base and generated creative records when the same file or variant is ingested multiple times (e.g., re-exports from a DAM or repeated generation runs).

Validate that every Generated/Modified Creative Asset references an existing Parent_Asset_ID or is explicitly marked as net-new, and that no “orphan” creatives exist without a valid audience, experience, or content link.

Timestamp Normalization

Convert all event timestamps (generation, safety checks, synthetic evaluations, activations) to UTC while retaining original local time where relevant as attributes.

Ensure consistent granularity for time fields (e.g., second-level for events, date-level for reporting snapshots) so evaluation and activation timelines can be compared reliably.

Audience & Segment Integrity

Maintain consistent Audience_Segment_ID definitions across segmentation pipelines,

synthetic audience construction, and activation mappings.

Validate that the segment attached to a creative matches the segment used to generate the synthetic audience evaluation and the segment targeted in downstream activation, so we don't mis-attribute performance.

Content & Experience Alignment

Enforce that every `Creative_Asset_ID` used in evaluation or activation has a valid `Content_ID` and `Experience_ID`, and that the placement's `Dimensions_Spec` matches the asset's format and size.

Prevent mismatches where creatives promote the wrong content or objective (e.g., engagement vs subscription upsell) by checking `Intended_Objective` against experience and segment metadata.

Policy, Rights & Safety Consistency

Ensure that every base and generated asset has an associated `Rights_Profile_ID`, and that allowed territories and channels align with the activation records.

Require that safety/brand checks exist and are up to date for any asset marked `Allowed_For_Synthetic_Testing` or `Allowed_For_Production_Use`, with no gaps where unreviewed assets enter testing or go live.

Synthetic Evaluation Integrity

Guarantee that each `Evaluation_Run_ID` references a valid `Synthetic_Audience_ID` and `Creative_Asset_ID`, and that the scenario context (placement, content, objective) is complete.

Enforce consistent scales and definitions for predicted metrics (click propensity, dwell time, subscription-start propensity, fatigue risk, brand fit) so scores are comparable across runs and segments.

Activation & Feedback Loop Quality

Verify that activated creatives (`Activation_Status = Live`) reference only assets approved by policy checks and, where required, evaluated by synthetic audiences.

Keep activation start/end timestamps aligned with evaluation timestamps and model versions so real-world performance can be accurately tied back to the correct creative, segment, and evaluation run.

Data Freshness & Latency SLAs

Define update cadences such as:

- Near real-time / hourly: new generated creatives, evaluations, and activation status changes.
- Daily: refreshed audience segments, content/offer metadata, and synthetic audience models or snapshots.

Implement alerts when key datasets (segments, creative catalogs, safety checks,

evaluation scores, activation mappings) fall behind their expected refresh schedule, so the creative command center doesn't recommend or activate stale or non-compliant assets.

Feature Engineering

Creative & Experience Features

Audience & Experience Features

- **Segment_Intent_Features:** indicators that a segment should see differentiated creative (e.g., `is_news_junkie`, `is_sports_superfan_ctv`, `is_churn_risk_family`).
- **Experience_Context_Features:** placement and objective descriptors (e.g., `placement_homepage_hero`, `surface_ctv_app`, `objective_subscription_upsell`).
- **Content_Offer_Fit_Features:** affinity and priority signals linking segment to content/offer (e.g., `sports_affinity_score`, `prior_upsell_response_score`, `content_priority_score`).

Creative Generation Features (Image & Video)

- **Prompt_Features:** structured fields derived from segment + experience (e.g., `prompt_tone_friendly`, `prompt_theme_family_movie_night`, `prompt_avoid_kids_faces`).
- **Model_Choice_Features:** which model/template/style to use (e.g., `model_family_id`, `brand_template_v3_flag`).
- **Rights_Constraint_Features:** what's allowed for this asset/context (e.g., `allow_ai_modification_flag`, `allowed_channels_mask`, `no_derivatives_flag`).

Quality, Policy & Safety Features

- **Creative_Quality_Features:** auto-evaluated quality metrics (e.g., `image_legibility_score`, `video_keyframe_quality_score`).
- **Safety_Brand_Features:** outputs from safety/brand checks (e.g., `safety_score`, `brand_compliance_score`, `contains_restricted_elements_flag`).
- **Rights_Usage_Features:** suitability for a specific use (e.g., `suitable_for_web_flag`, `suitable_for_email_flag`, `within_rights_window_flag`).

Synthetic Evaluation Features (Per Creative × Segment × Experience)

- **Engagement_Propensity_Features:** predicted engagement (e.g., `predicted_click_propensity`, `predicted_dwell_time`, `predicted_play_start_rate`).
- **Conversion_Propensity_Features:** downstream outcomes (e.g., `predicted_subscription_start_propensity`, `predicted_return_visit_7d`).

- **Fatigue_Risk_Features:** overexposure and annoyance risk (e.g., predicted_fatigue_risk, predicted_unsubscribe_risk).
- **Concept_Variant_Ranking_Features:** relative performance signals (e.g., concept_rank_within_segment, variant_rank_within_concept).

Activation & Outcome Features

- **Creative_Performance_Features:** real metrics per segment/placement (e.g., actual_ctr, actual_dwell_time_lift_vs_baseline, actual_sub_start_uplift).
- **Reuse_Generalization_Features:** how well a concept travels (e.g., cross_segment_success_rate, cross_surface_performance_std).
- **Operational_Features:** workflow friction (e.g., revision_count, approval_cycle_time_hours, policy_block_rate).

Intelligence Layers & Scores

1. **Creative Ranking Models**
 - **Objective:** Rank creatives for each segment and placement.
 - **Inputs:** Synthetic evaluation features, experience context, policy/rights features.
 - **Outputs:** Creative_Score, ranked creative list per segment × placement.
2. **Concept / Playbook Recommendation Models**
 - **Objective:** Recommend creative concepts and prompt patterns per segment and objective.
 - **Inputs:** Historical creative performance by concept and segment, audience & experience features.
 - **Outputs:** Recommended_Concepts, Concept_Suitability_Score.
3. **Quality & Safety Models**
 - **Objective:** Approve or block assets before testing/activation.
 - **Inputs:** Pixel/frame/text features, rights & policy metadata.
 - **Outputs:** Quality_Score, Safety_Score, Brand_Compliance_Score, allow/block flags.
4. **Synthetic Audience Response Models**
 - **Objective:** Predict segment response to creative in context.
 - **Inputs:** Creative features (incl. embeddings), synthetic audience profile, experience context.
 - **Outputs:** Predicted_Click_Propensity, Predicted_Dwell_Time, Predicted_Sub_Start_Propensity, Predicted_Fatigue_Risk.
5. **Learning & Calibration Models**
 - **Objective:** Align predictions and rankings with real outcomes over time.
 - **Inputs:** Real performance vs prior predictions and chosen creatives.

- Outputs: Calibration adjustments, Playbook_Health_Score, alerts on degrading concepts.

Target Personas

Product Manager, D2C Experience / Personalization

Owns the streaming or D2C app experience across web, mobile, and CTV, and is accountable for engagement, content discovery, and churn. They use the Creative Gen Command Center to decide which segments see which hero art, rails, and promo tiles in the app, and to test different creative concepts for driving starts and watch time on owned surfaces.

Growth / Lifecycle Lead (D2C subs)

Owns subscriber acquisition, upsell, winback, and retention journeys that touch the app (e.g., in-app interstitials, upsell modules, and coordinated email/push). They use the system to generate segment-specific creatives for app upsell surfaces and companion CRM (email/push) and to see which visuals best move users from “logged-in” to “paid,” or from “light viewer” to “engaged viewer.”

Design / Brand & Creative Ops for D2C

Owns the visual identity of the streaming product and ensures that all imagery in the app is on-brand and rights-compliant. They rely on the command center as a governed way to generate and adapt key visuals for app rails, heroes, and promos at scale, without violating IP, franchise, or regional guidelines.

Visualization

A Databricks-native Creative Gen Command Center for product, growth, and creative operations teams at a large media publisher with a D2C app. The app helps teams define audience segments, find base assets, generate and modify image/video creative, run policy and safety checks, shortlist the best options, evaluate the selected creatives with synthetic audiences, and activate approved creatives into owned experiences such as web, mobile, CTV, and downstream systems like Adobe Experience Manager or Adobe Target. Databricks Apps can support the custom application layer, while Genie provides embedded natural-language access to governed audience, creative, and performance data.

Use a **persistent left-side navigation** with these sections:

Home

Segments

Creatives

Evaluations

Activations

Ask

Top header

Use a sticky top header with only global elements:

App logo / name

Global search

Date range

Saved views

Notifications

User / team switcher (we'll start with just 1 view - Creative Ops)

Home

Purpose

Creative command center and daily summary view.

What it shows

- Priority segments needing new creative
- Creative generation runs in progress
- Blocked creative count by issue type
- Shortlists waiting for synthetic evaluation
- Evaluations awaiting decision
- Top-performing live activations

Description

This page should act like a daily briefing. It gives the user a quick view of what segments need attention, what creative work is underway, what has been blocked, what is waiting for evaluation, and what is already live. It should be high-level and scannable, helping the user decide where to go next.

Page interactions

- Open a segment needing new creative
- Review blocked creatives
- Open a shortlist awaiting evaluation
- Ask a question from the page itself

Segments

Purpose

Workspace for defining target audience segments and deciding where personalized creative should be used.

Segments list

What it shows

- Audience segment
- Segment type
- Size / reach
- Primary surfaces
- Primary objective
- Active creatives
- Last updated

Description

This is the planning view for audience-led creative personalization. It allows users to scan which segments are highest value, where differentiated creative is already live, and where the D2C experience still relies on generic assets.

Segment detail

Sections

- Overview
- Traits & Definition
- Eligible Surfaces
- Current Experience
- Creative History
- Next Best Action

Description

This page is the full working record for an audience segment. It combines who the segment is, where they appear in the D2C experience, what creative they are currently seeing, and what has worked before. It should help the user answer: who is this audience, where should we personalize, and what should we generate next?

Page interactions

- Create creative brief
- Select placements for this segment
- Launch generation workflow
- Ask a segment-specific question

Creatives

Purpose

Central workspace for searching base assets, generating 5 options, editing them, and selecting the top 2 for synthetic testing.

Creatives landing

What it shows

- Available base assets
- Recent generation runs
- Variants in review
- Blocked assets
- Human shortlists ready for evaluation

Sections

Library

Search the governed asset repository for images and videos by franchise, genre, topic, rights profile, approved channel, and prior usage.

Generate

The main creative-generation workspace. The user selects a segment, placement, base asset, and text prompt, and the system generates 5 creative options.

Edit

A lightweight modification workspace for resizing, cropping, reframing, inpainting, outpainting, and adapting creatives for homepage, app, email, and CTV placements.

Shortlist

After generation and any edits, the user reviews the 5 options and selects the best 2 to send to synthetic audience evaluation.

Review

Shows asset status (Draft, Blocked, Approved for Evaluation, Activated) and policy/safety results.

Description

This page helps users move from an audience need to a short list of viable creative options. The workflow should be simple: find a base asset, generate 5 variants, optionally edit them, and let the human pick the 2 strongest candidates. The goal is not to fully automate taste, but to accelerate ideation while keeping human judgment in the loop.

Blocked creatives handling

Do not show all blocked creatives on the homepage. Instead, Home shows a summary count by issue type, and clicking it opens the Creatives page filtered to blocked assets. The full blocked list is managed here with filters, search, and pagination.

Page interactions

- Search for base assets
- Generate 5 creative options
- Edit generated options
- Select top 2 for evaluation
- Review blocked or approved assets
- Ask a creative-specific question

Evaluations

Purpose

Workspace for synthetic audience scoring of the 2 human-selected options and final creative decision-making.

Evaluations list

What it shows

- Evaluation run
- Audience segment
- Placement
- Selected 2 creatives
- Status
- Top-ranked variant
- Predicted click propensity
- Predicted dwell-time impact
- Predicted subscription-start impact

Description

This is the decision layer for choosing which shortlisted creative should go live. It should show how the 2 human-selected variants performed against the synthetic audience for that same target segment, and help the user decide which one to activate.

Evaluation detail

Sections

- Overview

- Side-by-Side Comparison
- Synthetic Audience
- Score Breakdown
- Decision

Description

Overview summarizes the segment, placement, and selected variants. Side-by-Side Comparison presents the 2 chosen creatives next to each other. Synthetic Audience explains the modeled audience used for testing. Score Breakdown shows how each creative performed across propensity to click, dwell time, subscription-start propensity, fatigue risk, and brand fit. Decision is where the user picks the winner or sends both back for another generation round.

Page interactions

- Compare the 2 shortlisted creatives
- Approve the winning creative
- Send both back for revision or regeneration
- Ask an evaluation-specific question

Activations

Purpose

Workspace for pushing approved creatives into the publisher's D2C app and connected systems.

Activations list

What it shows

- Audience segment
- Placement / surface
- Winning creative
- Downstream system
- Status
- Start date
- End date
- Performance snapshot

Description

This page provides the operational view of where approved creatives have been activated. It helps users identify what is live across homepage, app, and CTV surfaces, which downstream systems have received the selected asset, and where creatives may need to be refreshed or replaced.

Activation detail

Sections

- Overview
- Delivery Context
- Downstream Sync
- Performance
- Refresh / Replace

Description

Overview gives the activation snapshot, including the segment, placement, selected creative, and destination system. Delivery Context shows how the creative appears in the D2C experience. Downstream Sync tracks pushes into Adobe Experience Manager, Adobe Target, CMS, or personalization tools. Performance compares observed results against synthetic predictions and generic baselines. Refresh / Replace helps users decide when a creative should be updated or retired.

Page interactions

- Activate creative
- Pause or replace activation
- Open downstream asset reference
- Export activation summary
- Ask an activation-specific question

Ask

Purpose

Natural-language interface across the application.

What it shows

- Prompt input
- Suggested prompts
- Recent questions
- Context selector
- Answer thread
- Supporting charts / tables

Description

Ask is the conversational layer for the app. Users can ask broad business questions or questions tied to a specific segment, creative set, shortlist, evaluation run, or activation. The experience should return both a direct answer and supporting evidence, with links into the relevant workflow when useful.

Page interactions

- Save query
- Open linked object
- Generate summary